UFUG 2106 Discrete Mathematics

Written Assignment 2

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Remark: There are 5 questions in this assignment. Please record how much time (in minutes) you spent on each question. This information is important to estimate the workload of this assignment and the difficulty of each question. It will NOT affect your grade.

Question	1	2	3	4	5
Time(min)					

1. Let the predicate variable domain be Z^+, the predicate $Odd(x)$ represent "x is odd", the predicate $Even(x)$ represent "x is even", the predicate $Prime(x)$ represent "x is prime", the predicate $Equal(x,y)$ represent "x equals y", the predicate $Greater(x, y)$ represent "x is greater than y". Please translate the following predicate statements into natural language statements and determine whether it is true or false.

(10 marks)

(a) $\exists x{\sim}Odd(x)$
(b) $\forall x(Prime(x) \rightarrow Greater(x, 1))$
(c) $\exists x \forall y(Greater(y, x))$
(d) $\forall x(Even(x) \rightarrow \exists y(Equal(x, y + 2)))$
(e) $\exists x \exists y(Prime(x) \wedge Prime(y) \wedge Equal(x + y, 10))$

(a) There exists a positive integer x such that x is not odd. Truth value: True; let $x=2$, $2 \in Z^+$ and 2 is not odd.
(b) For every positive integer x, if x is prime, then x is greater than 1. Truth value: True; by definition, any prime number is an integer greater than 1.
(c) There exists a positive integer x such that for every positive integer y, y is greater than x. Truth value: False; suppose such an x exists, take $y=x$, then the statement requires $x>x$, which is a contradiction.
(d) For every positive integer x, if x is even, then there exists a positive integer y such that $x=y+2$. Truth value: False. Counterexample: let $x=2$, then Even(2) is true. The equation $2=y+2$ implies $y=0$, but $0 \notin Z^+$. Hence no such y exists, so the implication fails for $x=2$.
(e) There exist positive integers x and y such that x and y are prime and $x+y=10$. Truth value: True; let $x=3$, $y=7$, then Prime(3) and Prime(7) are both true and $3+7=10$.

2. Let the predicate variable domain be Z^+, the predicate $Odd(x)$ represent "x is odd", the predicate $Even(x)$ represent "x is even", the predicate $Prime(x)$ represent "x is prime", the predicate $Equal(x,y)$ represent "x equals y", the predicate $Greater(x, y)$ represent "x is greater than y". Please translate the following natural language statements into predicate logic notation.

(10 marks)

(a) There exists an even number that is not prime.
(b) There exist two distinct odd positive integers whose sum is even.
(c) For any positive integers x, there exists a positive integer y such that $y > x$.
(d) There exists exactly one even prime number.
(e) For every prime number, there exists a larger prime number.

(a) $\exists x(Even(x) \wedge {\sim}Prime(x))$ (b) $\exists x \exists y(Odd(x) \wedge Odd(y) \wedge {\sim}Equal(x,y) \wedge Even(x+y))$ (c) $\forall x \exists y(Greater(y,x))$
(d) $\exists x(Even(x) \wedge Prime(x) \wedge \forall y((Even(y) \wedge Prime(y)) \rightarrow Equal(y,x)))$ (e) $\forall x(Prime(x) \rightarrow \exists y(Prime(y) \wedge Greater(y,x)))$

概念和写法

3. For each of the following predicate statement, please determine the scope, bound variables of each quantifier, and free variables (i.e., a variable that is not bound by any quantifier in the formula). (10 marks)

(a) $\exists x(P(x) \wedge Q(x)) \wedge \forall x R(x)$
(b) $\forall x(P(x) \rightarrow Q(x,y))$
(c) $\exists x(P(x) \wedge \forall y R(x,y))$
(d) $\forall x(P(x) \rightarrow Q(y)) \rightarrow \exists y(R(x) \wedge L(x,y,z))$
(e) $\forall x P(x,y,z) \vee (\exists u Q(x,u) \rightarrow \exists w Q(y,w))$

(a) scope of $\exists x$: $P(x) \wedge Q(x)$ scope of $\forall x$: $R(x)$
bound variables: The x in $P(x)$ and $Q(x)$ is bound by $\exists x$, the x in $R(x)$ is bound by $\forall x$
free variables: None

(b) scope of $\forall x$: $P(x) \rightarrow Q(x,y)$
bound variables: x free variables: y

(c) scope of $\exists x$: $P(x) \wedge \forall y R(x,y)$ scope of $\forall y$: $R(x,y)$
bound variables: x, y free variables: None

(d) scope of $\forall x$: $P(x) \rightarrow Q(y)$ scope of $\exists y$: $R(x) \wedge L(x,y,z)$
bound variables: The x in $P(x)$ is bound by $\forall x$, the y in $L(x,y,z)$ is bound by $\exists y$
free variables: x, y, z (specifically, the y in $Q(y)$ is free, the x is free in $R(x)$ and $L(x,y,z)$, the z is free in $L(x,y,z)$)

(e) scope of $\forall x$: $P(x,y,z)$ scope of $\exists u$: $Q(x,u)$ scope of $\exists w$: $Q(y,w)$
bound variables: The x in $P(x,y,z)$ is bound by $\forall x$; u is bound by $\exists u$; w is bound by $\exists w$
free variables: x, y, z (specifically, the y and z in $P(x,y,z)$ are free, the x in $Q(x,u)$ is free, the y in $Q(y,w)$ is free)

4. Please deduce the conclusion from the given premises. You should first translate the natural language statements into predicate statements, then use inference rules to deduce the conclusion. Show each step and provide the reason for each step. (8 marks)

let the domain be all people 先写domain

Premises:
(1) All mathematicians are logical thinkers.
(2) Some mathematicians are creative.
Conclusion: Some logical thinkers are creative.

let $M(x)$ be "x is a mathematician", $L(x)$ be "x is a logical thinker", $C(x)$ be "x is creative". Then

Premises: ① $\forall x(M(x) \rightarrow L(x))$ ② $\exists x(M(x) \wedge C(x))$

Conclusion: $\exists x(L(x) \wedge C(x))$

Step	Reason
1. $\exists x(M(x) \wedge C(x))$	Premise
2. $M(a) \wedge C(a)$	Existential instantiation from (1)
3. $M(a)$	Simplification from (2)
4. $\forall x(M(x) \rightarrow L(x))$	Premise
5. $M(a) \rightarrow L(a)$	Universal instantiation from (4)
6. $L(a)$	Modus ponens from (3) and (5)
7. $C(a)$	Simplification from (2)
8. $L(a) \wedge C(a)$	Conjunction from (6) and (7)
9. $\exists x(L(x) \wedge C(x))$	Existential generalization from (8)

5. Please deduce the conclusion from the given premises. You should first translate the natural language statements into predicate statements, then use inference rules to deduce the conclusion. Show each step and provide the reason for each step. (12 marks) bright 明亮 dull 暗淡

Premises:

(1) All birds that sing beautifully are brightly colored.

(2) No large birds primarily feed on honey.

(3) Birds that do not primarily feed on honey have dull colors.

Conclusion: No birds that sing beautifully are large birds.

Let the domain be all birds 先写domain

let $S(x)$ be "x sings beautifully", $B(x)$ be "x is brightly colored", $L(x)$ be "x is a large bird", $H(x)$ be "x primarily feeds on honey". $D(x)$ be "x has dull colors"

Then Premises: ① $\forall x(S(x) \to B(x))$ ② $\forall x(L(x) \to \sim H(x))$ ③ $\forall x(\sim H(x) \to D(x))$

Conclusion: $\forall x(S(x) \to \sim L(x))$ Background assumption: $\forall x(D(x) \to \sim B(x))$

Step	Reason
1. $\forall x(\sim H(x) \to D(x))$	Premise
2. $\sim H(a) \to D(a)$	Universal instantiation from (1)
3. $\forall x(D(x) \to \sim B(x))$	Background assumption
4. $D(a) \to \sim B(a)$	Universal instantiation from (3)
5. $\sim H(a) \to \sim B(a)$	Hypothetical syllogism using (2) and (4)
6. $B(a) \to H(a)$	Contrapositive of (5)
7. $\forall x(S(x) \to B(x))$	Premise
8. $S(a) \to B(a)$	Universal instantiation from (7)
9. $S(a) \to H(a)$	Hypothetical syllogism using (8) and (6)
10. $\forall x(L(x) \to \sim H(x))$	Premise
11. $L(a) \to \sim H(a)$	Universal instantiation from (10)
12. $H(a) \to \sim L(a)$	Contrapositive of (11)
13. $S(a) \to \sim L(a)$	Hypothetical syllogism using (9) and (12)
14. $\forall x(S(x) \to \sim L(x))$	Universal generalization from (13)